

# Block diagram

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## Mechatronics 3DX4

### Slides 3: Reduction of Multiple Subsystems

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Material based on lecture notes by P. Taylor and M. Lawford, and *Control Systems Engineering* by N. Nise.

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## Introduction

- ▶ So far we have represented systems as a single block (transfer function), with its inputs and outputs.
- ▶ Many systems are much more complicated and represented by many interconnected subsystems.
- ▶ As its straightforward to calculate the response of a single transfer function we want to be able to convert multiple

transfer function, we want to be able to convert multiple subsystems into an equivalent single transfer function.

- ▶ We will use **block diagram algebra** to do the reduction.
- ▶ This then allows us to apply the techniques we have already developed to the resulting single subsystem.

## Block Diagrams

- ▶ When interconnecting multiple subsystems, we need more elements than just a single block with inputs and outputs.
- ▶ We add the elements:
  - Summing Junctions:** they combine two or more signals, producing the algebraic sum as output.

**Pickoff Points:** it breaks input signal into multiple copies to be sent to different destinations.

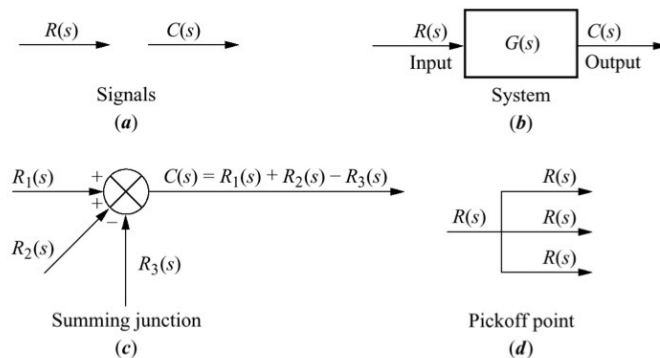


Figure 5.2

## Cascade Form

- ▶ First common interconnection method we look at is called the **cascade form**.
- ▶ Consists of two or more subsystems connected in a serial

fashion.

- Equivalent to a single block with transfer function equal to product of the individual block's transfer functions.

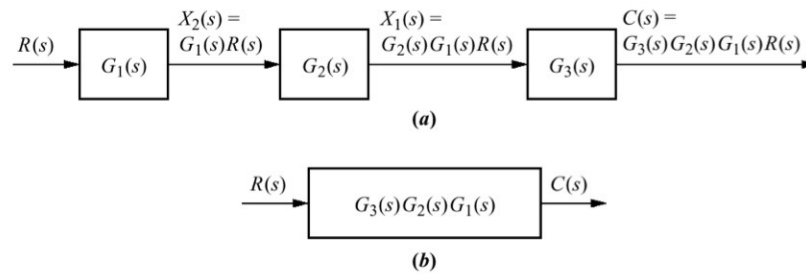


Figure 5.3.

## Loading in Cascaded Subsystems

- Formula for combining cascaded subsystems is invalid when a given subsystem **loads** its preceding subsystem.
- A given subsystem is not loaded by the next subsystem if its output is unchanged by connecting the following subsystem.

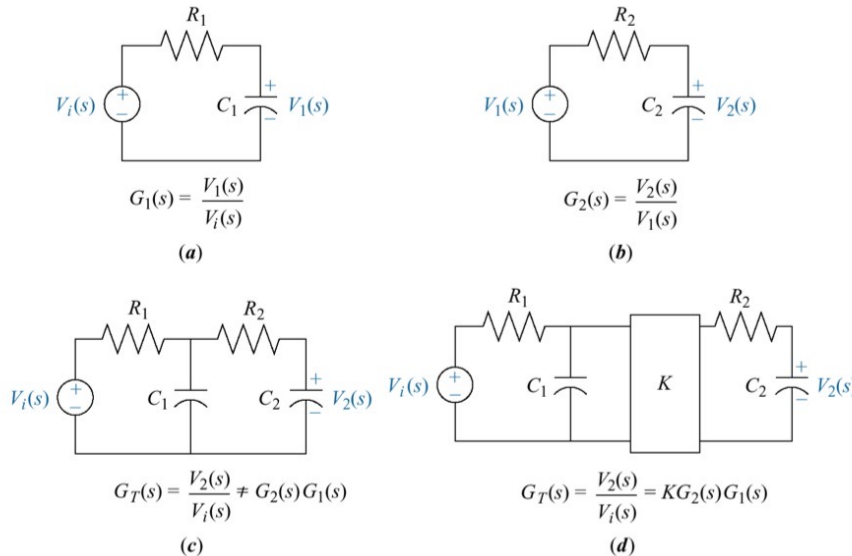


Figure 5.4.

## Parallel Form

- ▶ For subsystems connected in parallel:
  - ▶ They all have same input.
  - ▶ The output of the group is the sum of each individual subsystem's output.
- ▶ The equivalent transfer function is the sum of the individual transfer functions.

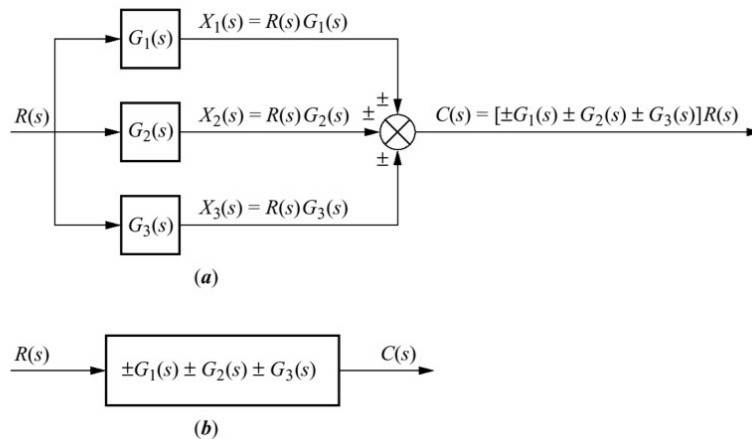


Figure 5.5.

## Feedback Form

- ▶ Feedback topology is basis of control systems theory.

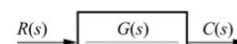
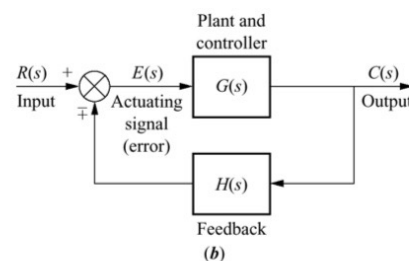
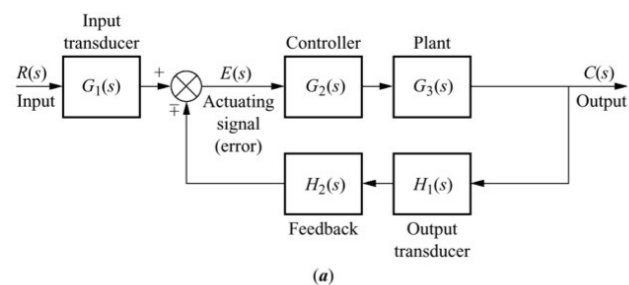
- ▶ In Simplified model (Fig. 5.6(b)), we see that:

$$E(s) = R(s) \mp C(s)H(s)$$

- ▶ We also see  $C(s) = E(s)G(s)$   
thus  $E(s) = \frac{C(s)}{G(s)}$ .

- ▶ Substituting in above gives:  
 $G_e(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$

- ▶ We call  $C(s)H(s)$  the error



we call  $G(s)H(s)$  the **open loop transfer function** or **loop gain**.

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Input  $\boxed{1 \pm G(s)H(s)}$  Output  
(c)

Figure 5.6.

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E是error signal，是想要的东西和反馈回来的差，根据正负号的不同其分别是正反馈和负反馈。

$$E(s) = R(s) \pm C(s)H(s) \quad (1)$$

$$C(s) = E(s)G(s) \quad (2)$$

sub (1) to (2)

$$C(s) = [R(s) \mp C(s)H(s)]G(s)$$

$$C(s) = R(s)G(s) \mp C(s)H(s)G(s)$$

$$C(s) \pm C(s)H(s)G(s) = R(s)G(s)$$

$$C(s)[1 \pm H(s)G(s)] = R(s)G(s)$$

$$C(s) = \frac{R(s)G(s)}{1 \pm H(s)G(s)}$$

↓  
We want + at here to make it stable

so, for  $\mp$  feed back get  $\pm$  at there, that's the reason why we use negative feed back.

我们通常用负反馈的原因是正反馈会导致pole变成 $1-H(s)G(s)$ ，这会让pole在RHP变得不稳定。

## Moving Blocks to Create Familiar Forms

- ▶ Have examined three different topologies so far.
- ▶ In physical systems, we will find them combined into complex arrangements.
- ▶ Recognizing these structures will be key to reducing more complex systems to a single transfer function.
- ▶ Unfortunately, these forms may be present, but not always obvious.
- ▶ We will learn how to move blocks forward or backwards past summing junctions and pickoff points.

## Moving Blocks Through Summing Junctions

- ▶ Top figure shows the equivalent diagram when block moved to the left of junction.
- ▶ Can see they are equivalent by noting that on left,  

$$C(s) = [R(s) \mp X(s)]G(s) = R(s)G(s) \mp X(s)G(s).$$
- ▶ Bottom figure shows equivalent system when moving block to the right of junction.
- ▶ Can see equivalent since on right,

$$\begin{aligned} C(s) &= \left[ R(s) \mp \frac{X(s)}{G(s)} \right] G(s) \\ &= R(s)G(s) \mp X(s) \end{aligned}$$

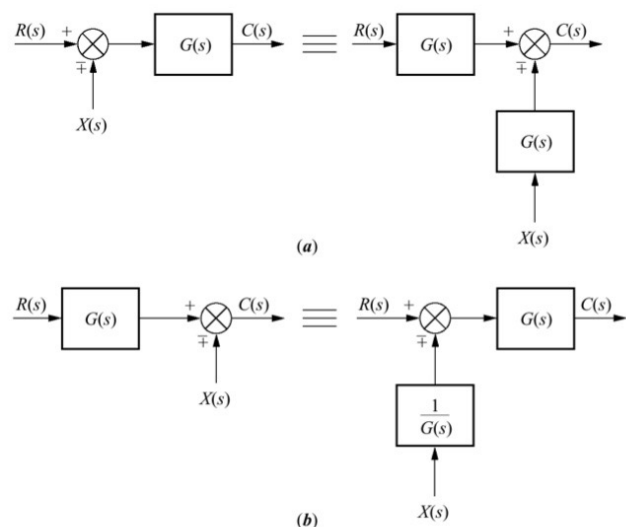


Figure 5.7.

**注意：移动只能移动起点，不能乱移动终点，即连接到加法器/减法器的输出箭头！！！！**

**可以移动加法器左右的block，但是不能动加法器上的信号。**

## Moving Blocks Through Pickoff Points

- ▶ Top figure shows the equivalent diagram when block moved to the left of pickoff point.
- ▶ Bottom figure shows equivalent system when moving block to the right of pickoff point.

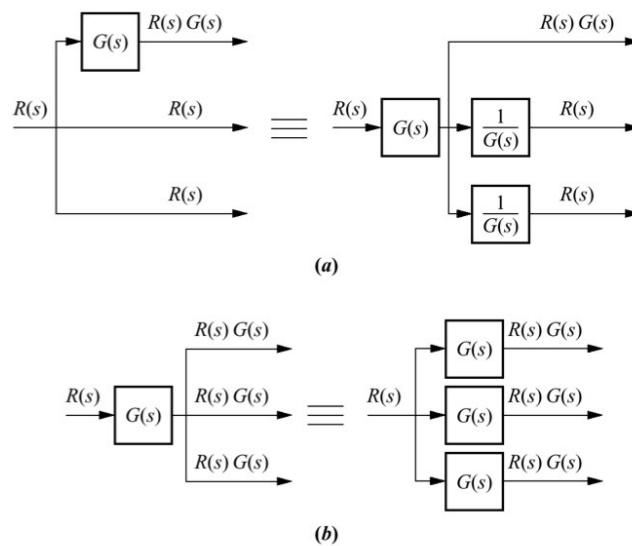
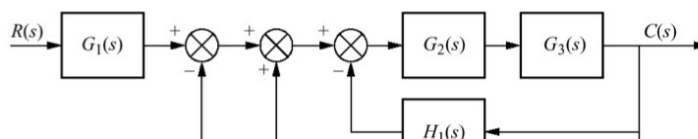


Figure 5.8.

## Reduction Via Familiar Forms eg.

- ▶ Reduce block diagram to a single transfer function.



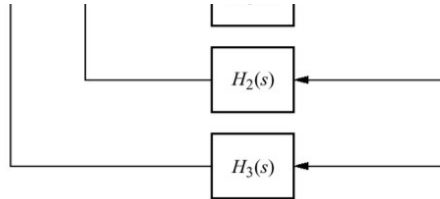


Figure 5.9.

- We start by noting that the three summations are just doing algebraical sums and can be combined

## Reduction Via Familiar Forms eg. - II

- Combining the summations gives Fig. (a).
- Applying parallel and cascade rule gives Fig. (b).
- Applying feedback rule, followed by cascade rule to combine with  $G_1(s)$ , gives Fig. (c).

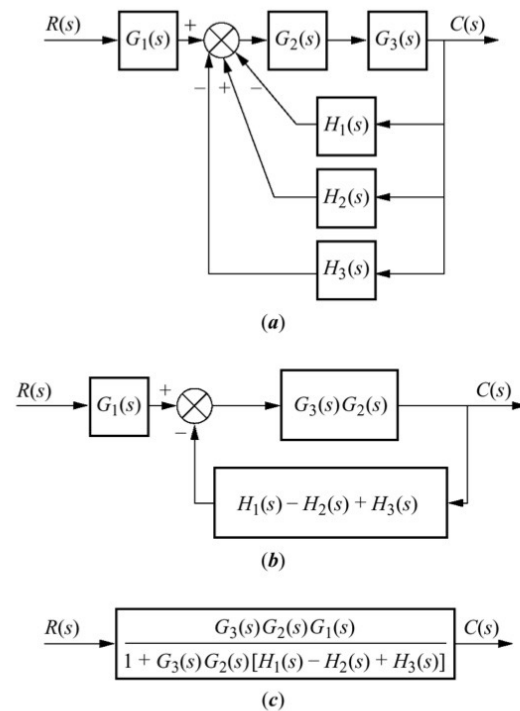


Figure 5.10.

## Reduction by Moving Block eg.

- Reduce block diagram to a single transfer function

- Reduce block diagram to a single transfer function.

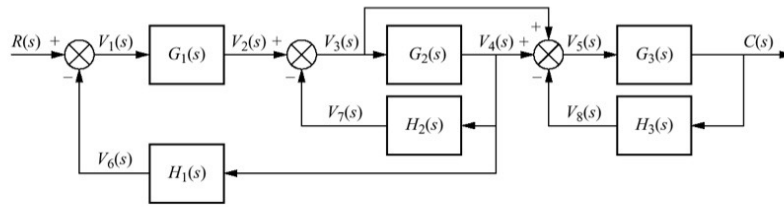


Figure 5.11.

1. Move  $G_2$  to left of pickoff point creating parallel form.
2. Reduce feedback system ( $G_3, H_3$ ).

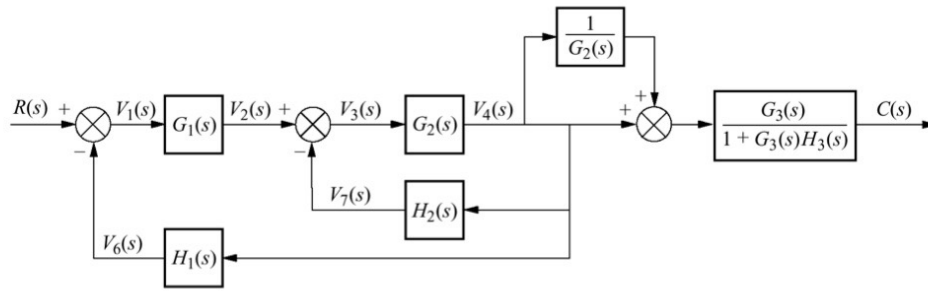


Figure 5.12.

## Reduction by Moving Block eg. - II

3. Reduce parallel form containing  $\frac{1}{G_2(s)}$  and unity.
4. Push  $G_1(s)$  to the right past summing junction. Creates parallel form ( $H_1$  and  $[\frac{1}{G_1}, H_2]$ ).
5. Combine serial forms ( $G_1, G_2$ ) and  $(\frac{1}{G_1}, H_2)$ .

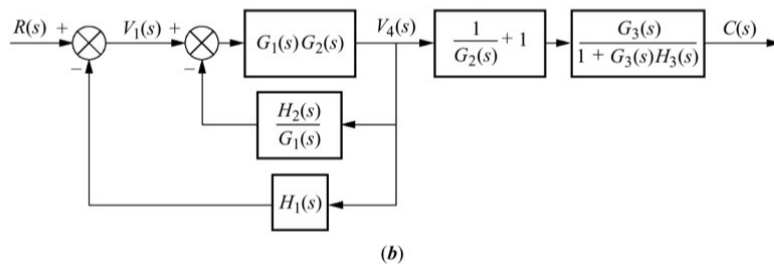
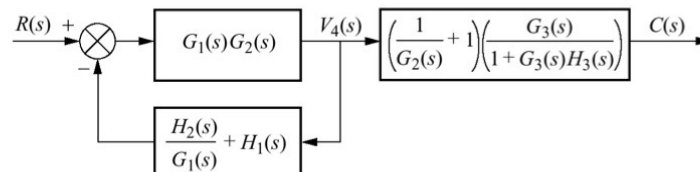


Figure 5.12.

6. Collapse summing junctions, and combine parallel form.
7. Combine serial form on right.



## Reduction by Moving Block eg. - III

### 8. Collapse feedback form.

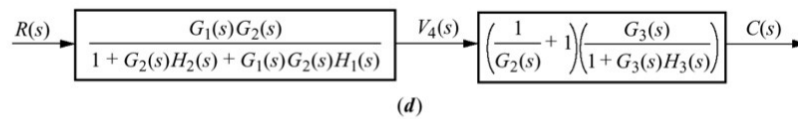


Figure 5.12.

### 9. Combine the two cascade blocks.

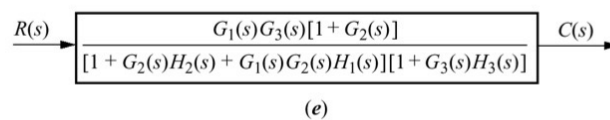
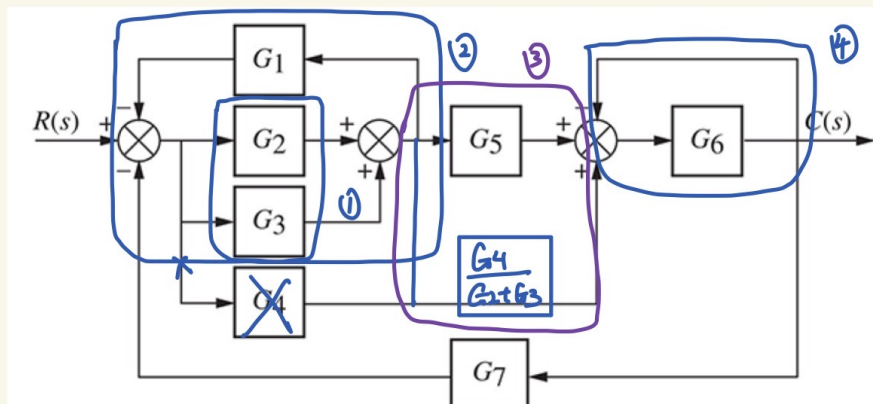


Figure 5.12.

25.



1)  $a = b + c$       2)  $a = \frac{G_2 + G_3}{G_2 + G_3}$

$$0 = G_2 + G_3 \quad 0 = G_2 + G_1(G_2 + G_3) + 1$$

$$\textcircled{3} g_3 = G_5 + \frac{G_4}{G_2 + G_3} \quad \textcircled{4} g_4 = \frac{G_6}{G_6 + 1}$$

$$\therefore G_o = g_2 g_3 g_4 \quad \therefore T = \frac{G_o}{1 + G_o \cdot G_7}$$

```
syms G1 G2 G3 G4 G5 G6 G7
g1=G2+G3;
g2=g1/(1+g1*G1);
disp('g2=');
disp(g2);
g3=G5+G4/(G2+G3);
disp('g3=');
disp(g3);
g4=G6/(1+G6);
disp('g4=');
disp(g4);
G_o=g2*g3*g4;
disp('G_o=');
disp(G_o);
T=G_o/(1+G_o*G7);
disp('T=');
disp(simplifyFraction(T));
T=
```